Emre Yavuz

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Github Linkedin Portfolio

ABOUT ME

I am a graduate of Gebze Technical University, Department of Computer Engineering. I have developed mobile and web applications, worked on backend architectures, and written hardware-level code for embedded systems. I have also carried out projects in artificial intelligence, computer vision, real-time operating systems, and large language models. As a disciplined, solution-oriented engineer who is open to learning, I can quickly add value to your team and projects.

Gebze Technical University – B.Sc. in Computer Engineering in English (2020–2025)

EXPERIENCES

TÜBİTAK BİLGEM FPGA Design Internship **07/2025 - 08/2025**

- Designed and verified a UART module in **VHDL** with oversampling, metastability protection, and error recovery logic.
- Integrated the **UART IP** into a MicroBlaze embedded system in AXI4-Lite and implemented **hardware–software communication** in **C**.
- Conducted timing simulations, self-checking testbenches, and verification for real-time reliability.
- Gained hands-on experience in **Vivado**, **FPGA** design flow, and **embedded C** firmware development.
- Improved **system synchronization** and signal integrity by **30%**, ensuring robust real-time performance.

ORTEM Electronics Internship **01/2025 - 02/2025**

- Developed a real-time **Flutter dashboard** to process **UART/CAN** data through **RESTful APIs** and serial communication .
- Built modular architecture for **synchronized** vehicle parameters, integrating backend and embedded for stable **real-time** performance.
- Implemented **data buffering, reconnection logic**, and **error-handling** mechanisms, reducing message loss by **40%**.
- Strengthened serial communication reliability by improving message sequencing and automatic reconnection in unstable links.
- Designed UI components (speedometer, odometer, gear selector) synchronized with **CAN** data for automotive applications.

TECHNICAL SKILLS AND TOOLS

Programming Languages	• Java, C/C++, Python,Verilog/VHDL
Frameworks & Libraries	• Node.js (Express)
Backend	• PostgreSQL,Firebase, RESTful APIs
DevOps & Tools	• Git, GitHub Actions, Docker, CI/CD
Others	• Linux, Clean Architecture, OOP / SOLID, Computer Vision, Operating Systems

PROJECTS

LLM-Based Cybersecurity Maturity Assessment Tool:

Developed a **Node.js (Express)** and **React**-based system to evaluate organizations **cybersecurity** maturity levels. Implemented dynamic questionnaire flow using **OpenAI** GPT and **LangChain**, and managed data with a **PostgreSQL** and **Docker**-based architecture. Integrated **Redmine API** and automated PDF reporting, reducing assessment time by **60%.**

Smart Environmental Monitoring and Control System:

Designed a system that processes multi-sensor data (light, heat etc.) and enables communication between **FPGA** and microcontroller via **I²C**, **SPI**, and **UART** protocols. Built a **real-time** visual and auditory alert system using LEDs, and a buzzer.

Pseudo-Labeled Keypoint Detection and Feature Extraction:

Developed a **deep learning**–based keypoint detection and **feature extraction model** using **pseudo-labeled** data. Employed **convolutional neural networks** and classical **computer vision** techniques for feature matching and stereo correspondence tasks.

Operating System and CPU Simulator (Assembly, Python):

I developed a **CPU simulator** and an **operating system kernel** running on a **custom ISA**. On CPU side, I implemented memory management, user/kernel modes, and system call mechanisms. On OS side, I designed TCB structures to manage **thread** states, with a round-robin **scheduler.** The system is capable of managing 10 threads and running complex algorithms **concurrently**.

Click for more project examples↗